# Classification : Decision Tree (DT)

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# Outline

- What is decision tree (DT) algorithm
- Why we need DT
- The Pros and Cons of DT
- Information Theory
- Some Issues in DT
- Assignment II



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# Decision Tree (DT)

- *Decision trees:* splitting datasets one features at a time.
- The decision tree is one of *commonly* used the *most* classification technique.
  - It has a *decision blocks* (*rectangles*).
  - *Termination block* (*ovals*).
  - The *right* and *left* arrows called *branches*.
- The *kNN algorithm* can do a grate job of *classification*, but it didn't lead to any *major insight* about the data.



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# Decision Tree (DT)

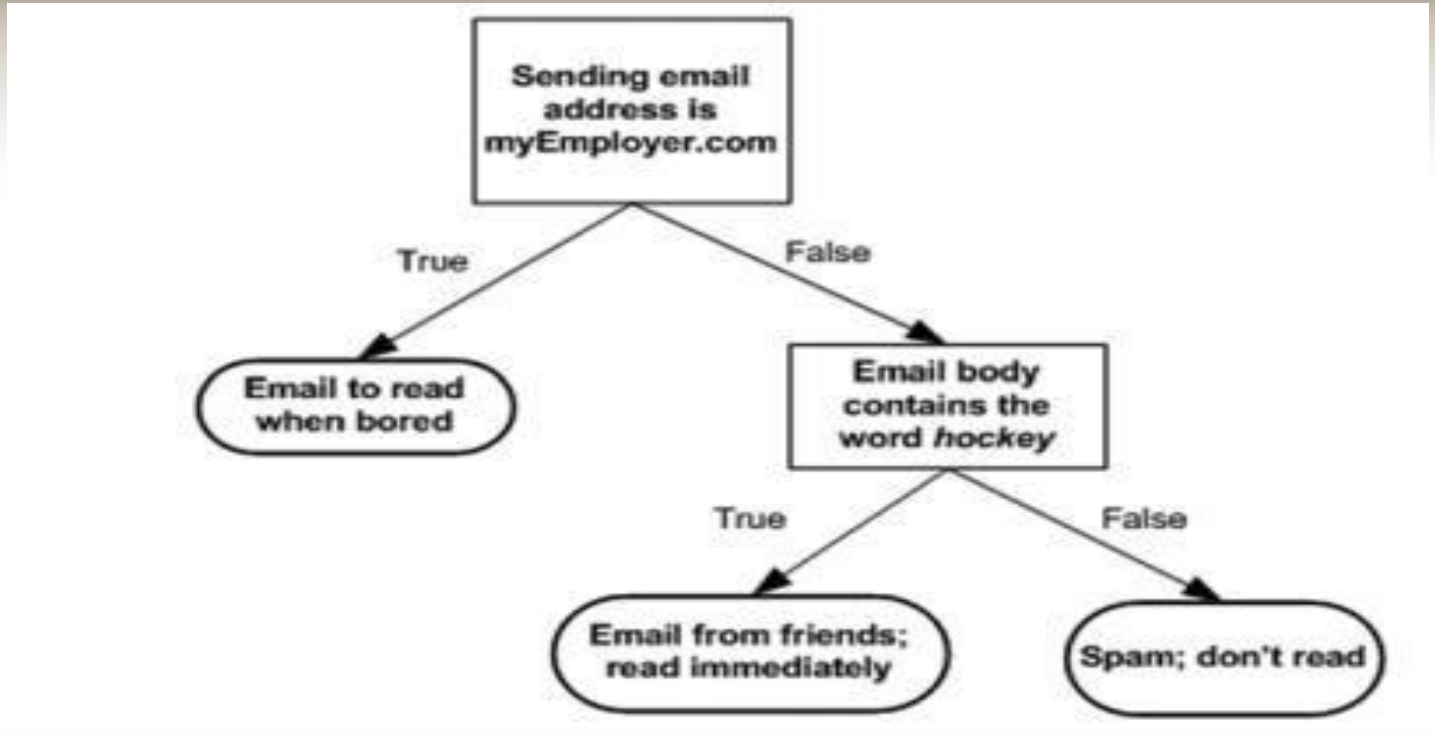


*Figure 1*: A decision Tree



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# Decision Tree (DT)

- The best part of the DT algorithm is (decision tree) humans can that *easily understand the data*:
- The DT algorithm:
  - Takes a *set of data*. (training examples)
  - Build a *decision tree (model)*, and *draw* it.
- It can also be re-represented as *sets of if-then rules* to improve human readability.
- The DT does a grate job of *distilling data* into *knowledge*.
  - Takes a *set of unfamiliar data* and extract a *set of rules*.
  - DT is often used in *expert system* development.



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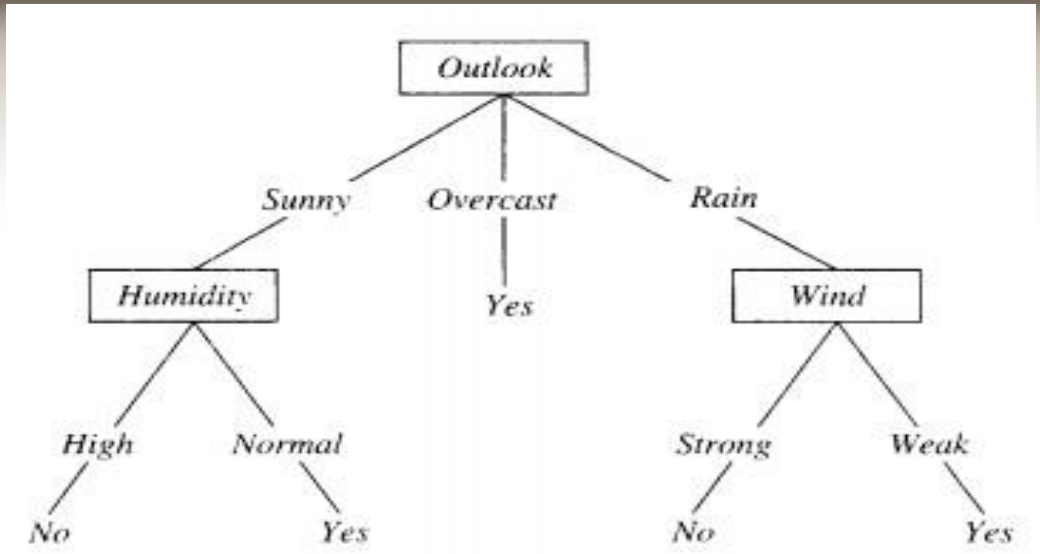
# Decision Tree (DT)




- The DT can be expressed using the following expression:
  - (Outlook = *Sunny* ∧ Humidity = *Normal*) → Yes
  - ∨ (Outlook = *Overcast*) → Yes
  - ∨ (Outlook = *Rain* ∧ Wind = *Weak*) → Yes



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# Decision Tree (DT)

- The pros and cons of DT:
- *Pros*:
  - Computationally cheap to use,
  - Easy for humans to understand the learned results,
  - Missing values OK (robust to errors),
  - Can deal with irrelevant features.
- *Cons*:
  - Prone to overfitting.
- *Work with*: *Numeric* values, *nominal* values.



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# Overfitting

- Overfitting refers to a model that models the training data too well.
- Overfitting happens when a model learns the detail and noise in the training data to the extent that it negatively impacts the performance of the model on new data.
- This means that the noise or random fluctuations in the training data is picked up and learned as concepts by the model.
- The problem is that these concepts do not apply to new data and negatively impact the models ability to generalize.



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# Underfitting

- Underfitting refers to a model that can neither model the training data nor generalize to new data.
- An underfit machine learning model is not a suitable model and will be obvious as it will have poor performance on the training data.
- Underfitting is often not easy to detect given a good performance metric.
- The solution is to move on and try alternate machine learning algorithms. Nevertheless, it does provide a good contrast to the problem of overfitting.



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# Decision Tree (DT)

- Appropriate problems for DT learning:
  - Instance are represented by *attribute-value pairs* (fixed set of attributes and their values),
  - The *target function* has discrete output values,
  - *Disjunctive* descriptions may be required,
  - The training data may *contain errors*,
  - The training data may contain *missing attribute values*.



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# Decision Tree (DT)

- The mathematics that is used by DT to *split the dataset* is called *information theory*:
- The first decision, you need to make is:
  - Which *feature* shall be used to split the data.
  - You need to try *every feature and measure* which split will give the *best result*.
  - Then split the dataset into *subsets*.
  - The *subsets* will then traverse down the branches of the decision node.
  - **If** the data on thebranch is the *same, stop*;

    **else** repeat the *splitting*.



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# Decision Tree (DT)

```
Check if every item in the dataset is in the same class:
    If so return the class label
    Else
        find the best feature to split the data
        split the dataset
        create a branch node
            for each split
                call createBranch and add the result to the branch node
        return branch node
```

*Figure 2*: Pseudo-code for the splitting function



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# Decision Tree (DT)

- General approach to *decision trees*:
  - *Collect*: Any method.
  - *Prepare*: ID3 algorithm works only on *nominal values*, so any *continues values* will need to be *quantized*.
  - *Analyze*: Any method. You should visually inspect the tree after it is build.
  - *Train*: Construct a tree data structure.
  - *Test*: Calculate the error rate with the learned tree.
  - *Use*: This can be used in any supervised learning task. Often, to better understand the data.



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# Decision Tree (DT)

- We would like to classify the following animals into *two classes*:
  - Fish and not Fish

| | Can survive without coming to surface? | Has flippers? | Fish? |
|---|---|---|---|
| 1 | Yes | Yes | Yes |
| 2 | Yes | Yes | Yes |
| 3 | Yes | No | No |
| 4 | No | Yes | No |
| 5 | No | Yes | No |

*Table 1*: Marine animal data



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# Decision Tree (DT)

- Need to decide whether we should split the data based on the *first feature* or the *second feature*:
  - To make *more organize* the *unorganized* data.
  - One way to do this is *to measure* the information.
  - Measure the information *before* and *after* the split.

- *Information theory* is a branch of science that is concerned with *quantifying information*.
- The *change in information* before and after the split is known as the *information gain*.



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# Decision Tree (DT)

- The split with the *highest information gain* is the best option.
  - The *measure of information* of a set is known as the *Shannon entropy* or *entropy*.
  - One way to do this is to measure the *information*.

- The change in information before and after the split is known as the *information gain*.



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# Decision Tree (DT)

- To *calculate entropy*, you need the expected value of all the information of all possible values of *our class*.
- This is given by:

$$H = -\sum_{i=1}^{n} p(x_i) \log_2 p(x_i)$$

- Where *n* is the number of *classes*:



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# Decision Tree (DT)

- The higher the *entropy*, the *more mixed* up the data.
- Another common measure of disorder in a set is the *Gini impurity*.
  - Which is the *probability* of choosing an item from the set and the probability of that *data item being misclassified*.
  - Calculate the *shannon entropy* of a dataset.
  - Dataset splitting on a given feature.
  - Choosing the *best feature* to split on.



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# Decision Tree (DT)

- Recursively building the tree.
  - Start with dataset and split it based on the *best attribute* to split.
  - The data will *traverse down the branches* of the tree to another node.
  - This node will then split the *data again* (recursively)
  - Stop under the following conditions: *run out of attributes* or if the instances in a branch are the *same class*.

- **ID3** stands for Iterative Dichotomizer 3 and is named such because the **algorithm** iteratively (repeatedly) dichotomizes(divides) features into two or more groups



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# Decision Tree (DT)

| | No surfacing | Flippers? | Fish? |
|---|---|---|---|
| 1. | Yes | Yes | Yes |
| 2. | Yes | Yes | Yes |
| 3. | Yes | No | No |
| 4. | No | Yes | No |
| 5. | No | Yes | No |

*Table 2:* Example training sets



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# Decision Tree (DT)

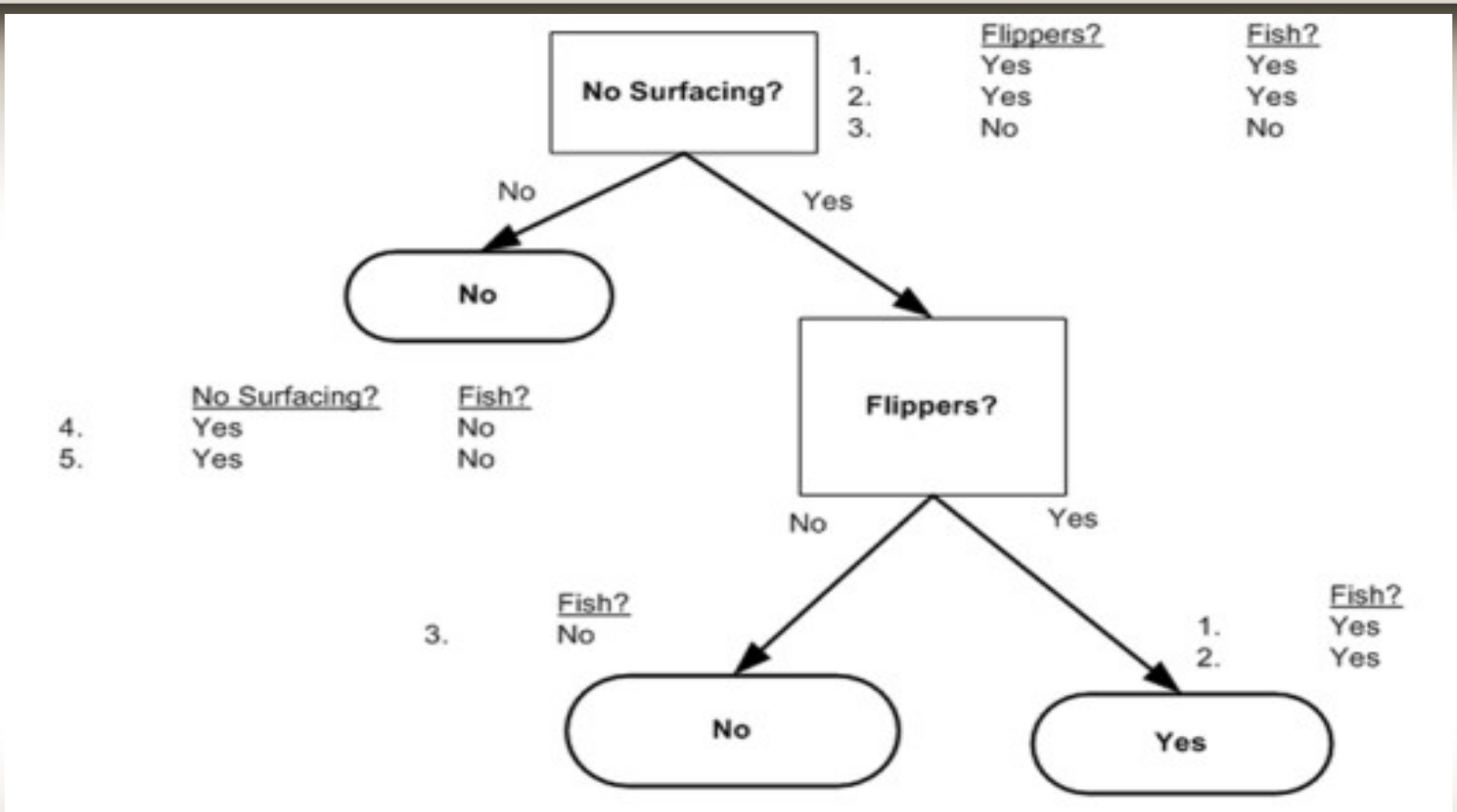


*Figure 3:* Data path while splitting



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# Decision Tree (DT)

- ID3 uses the *information gain* measure to select among the *candidate attributes*.
  - Start with dataset and split it based on the *best attribute* to split.
  - Given a collection S, containing *positive* and *negative* examples of *some target*.
  - The entropy of S relative to this *Boolean classification* is:
    - **Entropy(S) = $-p_1 \log_2 p_1 + - p_2 \log_2 p_2$**

Where Entropy refer: measure of randomness in the data. The higher the entropy, the harder its to draw any conclusions from the data

Entropy in a DT determines how DT chooses to split data



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# Decision Tree (DT)

- Example:
  - The target attribute is *PlayTennis*. (yes/no)

| Day | Outlook | Temperature | Humidity | Wind | PlayTennis |
|---|---|---|---|---|---|
| D1 | Sunny | Hot | High | Weak | No |
| D2 | Sunny | Hot | High | Strong | No |
| D3 | Overcast | Hot | High | Weak | Yes |
| D4 | Rain | Mild | High | Weak | Yes |
| D5 | Rain | Cool | Normal | Weak | Yes |
| D6 | Rain | Cool | Normal | Strong | No |
| D7 | Overcast | Cool | Normal | Strong | Yes |
| D8 | Sunny | Mild | High | Weak | No |
| D9 | Sunny | Cool | Normal | Weak | Yes |
| D10 | Rain | Mild | Normal | Weak | Yes |
| D11 | Sunny | Mild | Normal | Strong | Yes |
| D12 | Overcast | Mild | High | Strong | Yes |
| D13 | Overcast | Hot | Normal | Weak | Yes |
| D14 | Rain | Mild | High | Strong | No |

*Table 3:* Example training sets



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# Decision Tree (DT)

- Suppose *S* is a collection of *14 examples* of some Boolean concept, including *9 positive* and *5 negative* examples.
- Then the entropy of *S* relative to *Boolean classification* is:

$$Entropy([9+, 5-]) = -(9/14)\log_2(9/14) - (5/14)\log_2(5/14)$$
$$= 0.940$$



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# Decision Tree (DT)

- Note that the entropy is *0* if all members of *S* belong to the *same class*.
- *For example*: if all the members are *positive* (p+ = 1), then (p- =0).
  - Entropy (s) = $-1.\log_2(1) - 0.\log_2(0) = 0$
- Note the entropy is *one (1)* when the collection contain an equal number of *positive* and *negative* examples.
- If the collection contain *unequal* number of *positive* and *negative* the entropy is b/n *0 and 1*.



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# Decision Tree (DT)

- Suppose S is a collection of training-example days described by attributes Wind. (*weak*, *strong*)
- The *information gain* is the measure used by *ID3* to select the best attribute at each step in growing the tree.

$$
\begin{aligned}
Values(Wind) &= Weak, Strong \\
S &= [9+, 5-] \\
S_{Weak} &\leftarrow [6+, 2-] \\
S_{Strong} &\leftarrow [3+, 3-] \\
Gain(S, Wind) &= Entropy(S) - \sum_{v \in \{Weak, Strong\}} \frac{|S_v|}{|S|} Entropy(S_v) \\
&= Entropy(S) - (8/14) Entropy(S_{Weak}) \\
&\quad - (6/14) Entropy(S_{Strong}) \\
&= 0.940 - (8/14)0.811 - (6/14)1.00 \\
&= 0.048
\end{aligned}
$$



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# Entropy of Wind

- Values(wind)=weak, strong
- S=[9+,5-] $S_{weak}$□[6+,2-] $S_{strong}$□[3+,3-]
- Entropy(weak)= $6/8\log_2 6/8 - 2/8\log_2 2/8$

$= (0.75)\log_2(0.75) - (0.25)\log_2(0.25)$

$= 0.311+0.5 = 0.811$

- Entropy(Strong)= $3/6\log_2 3/6 - 3/6\log_2 3/6$

$= (0.5)\log_2(0.5) - (0.5)\log_2(0.5)$

$= 0.5+0.5 = 1$

Gain(S, Wind)=Entropy(S) – (8/14)Entropy(weak) – (6/14)Entropy(strong)

$= 0.940 - (8/14)0.811 - (6/14)1$

$= 0.048$



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# Entropy of Humidity

- Values(Humidity)=High, Normal
- S=[9+,5-] $S_{High}$[3+,4-] $S_{Normal}$[6+,1-]
- Entropy(High)= $3/7\log_2 3/7 - 4/7\log_2 4/7$

$= -(0.429)\log_2(0.429) - (-0.571)\log_2(0.571)$

$= 0.524+0.461$ $= 0.985$

- Entropy(Strong)= $6/7\log_2 6/7 - 1/7\log_2 1/7$

$= (0.857)\log_2(0.857) - (0.143)\log_2(0.143)$

$= 0.191+0.401$ $= 0.592$

Gain(S, Humidity)=Entropy(S) – (7/14)Entropy(high) – (7/14)Entropy(normal)

$= 0.940 - (7/14)0.985 - (7/14)0.592$

$= 0.151$



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# Entropy of Temperature

- Values(Temperature)=Hot, Mild, Cool

S=[9+,5-] $S_{Hot}$□[2+,2-] $S_{Mild}$□[4+,2-] $S_{Cool}$□[3+,1-]

- Entropy(Hot)= $2/4\log_2 2/4 - 2/4\log_2 2/4$

$= -(0.5)\log_2(0.5) - (-0.5)\log_2(0.5)$

$= 0.5+0.5 \quad = 1$

- Entropy(Mild)= $4/6\log_2 4/6 - 2/6\log_2 2/6$

$= (0.667)\log_2(0.667) - (0.333)\log_2(0.333)$

$= 0.390+0.528 \quad = 0.918$

- Entropy(Cool)= $3/4\log_2 3/4 - 1/4\log_2 1/4$

$= (0.75)\log_2(0.75) - (0.25)\log_2(0.25)$

$= 0.311+0.5 \quad = 0.811$

Gain(S, Humidity)=Entropy(S) – (4/14)Entropy(high)–(6/14)Entropy(normal) – (4/14)Entropy(normal)

$= 0.940 - (4/14)1 - (6/14)0.918 - (4/14)0.811$

$= 0.029$



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# Entropy of Outlook

- Values(Outlook)=Sunny, Overcast, Rain

S=[9+,5-] $S_{Sunny}$[2+,3-] $S_{Overcast}$[4+,0-] $S_{Rain}$[3+,2-]

- Entropy(Sunny)= $2/5\log_2 2/5 - 3/5\log_2 3/5$

$= -(0.4)\log_2(0.4) - (-0.6)\log_2(0.6)$

$= 0.529+0.442 \quad = 0.971$

- Entropy(Overcast)= $4/4\log_2 4/4 - 0/4\log_2 0/4$

$= (1)\log_2(1) - (0)\log_2(0)$

$= 0+0 \quad = 0$

- Entropy(Rain)= $3/5\log_2 3/5 - 2/5\log_2 2/5$

$= (0.6)\log_2(0.6) - (0.4)\log_2(0.4)$

$= 0.442+0.529 \quad = 0.971$

Gain(S, Outlook)=Entropy(S) – (4/14)Entropy(sunny)–(6/14)Entropy(overcast) – (4/14)Entropy(rain)

$= 0.940 - (5/14)0.971 - (4/14)0 - (5/14)0.971$

$= 0.246$



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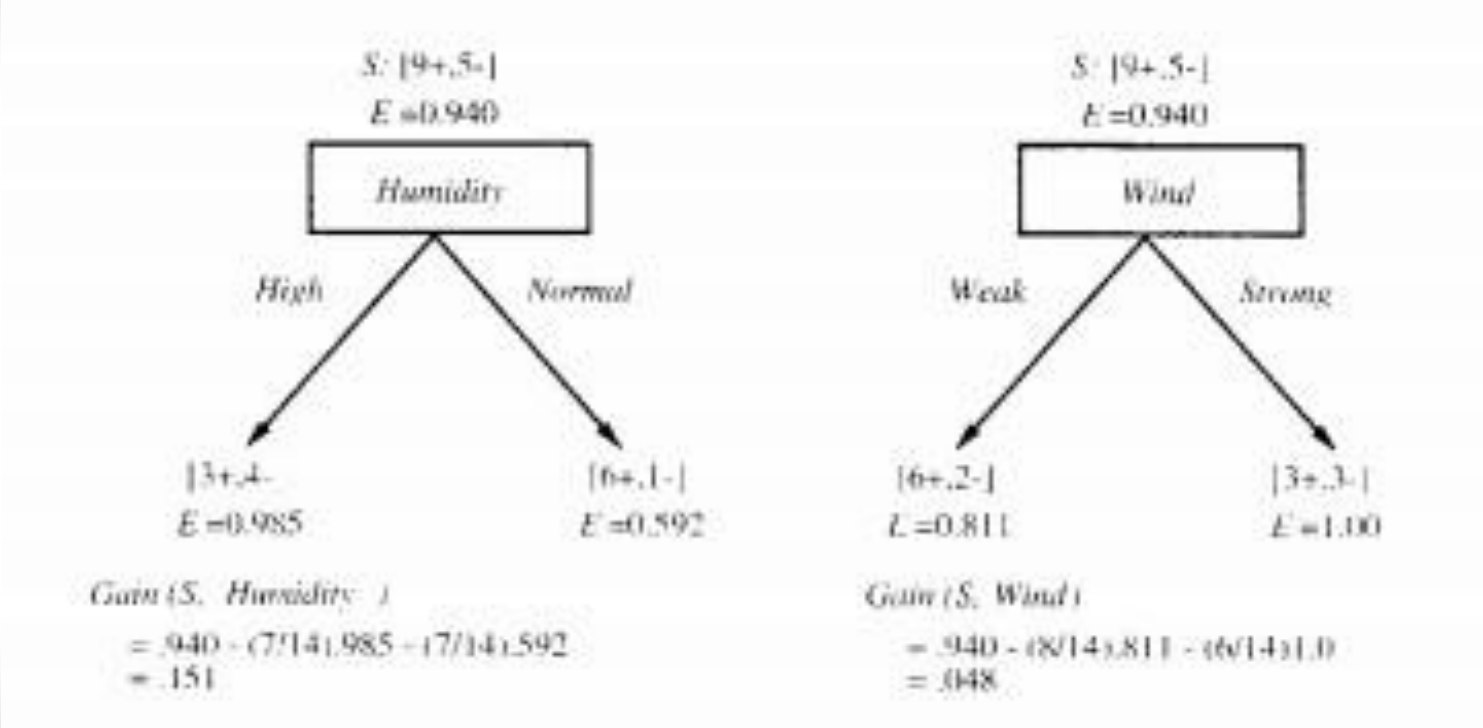
# Decision Tree (DT)

❖ Information gain of the two attributes: *Humidity* and *Wind*



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# Decision Tree (DT)

- Example:
  - ID3 will determines the *information gain* for each attribute.

(*Outlook*, *Temperature*, *Humidity* and *Wind*)

  - Then select the one with the *highest information gain*.
  - The information gain values for all *four attributes are*:
    - Gain(S, *Outlook*) = 0.246
    - Gain(S, *Humidity*) = 0.151
    - Gain(S, *Wind*) = 0.048
    - Gain(S, *Temperature*) = 0.029

  - *Outlook* provides *grater information gain* than the other.



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# Decision Tree (DT)

- Example:
  - According to the information gain measure, the Outlook attribute selected as the *root node*.
  - Branches are created below the root for each of its *possible values*. (*Sunny*, *Overcast*, and *Rain*)



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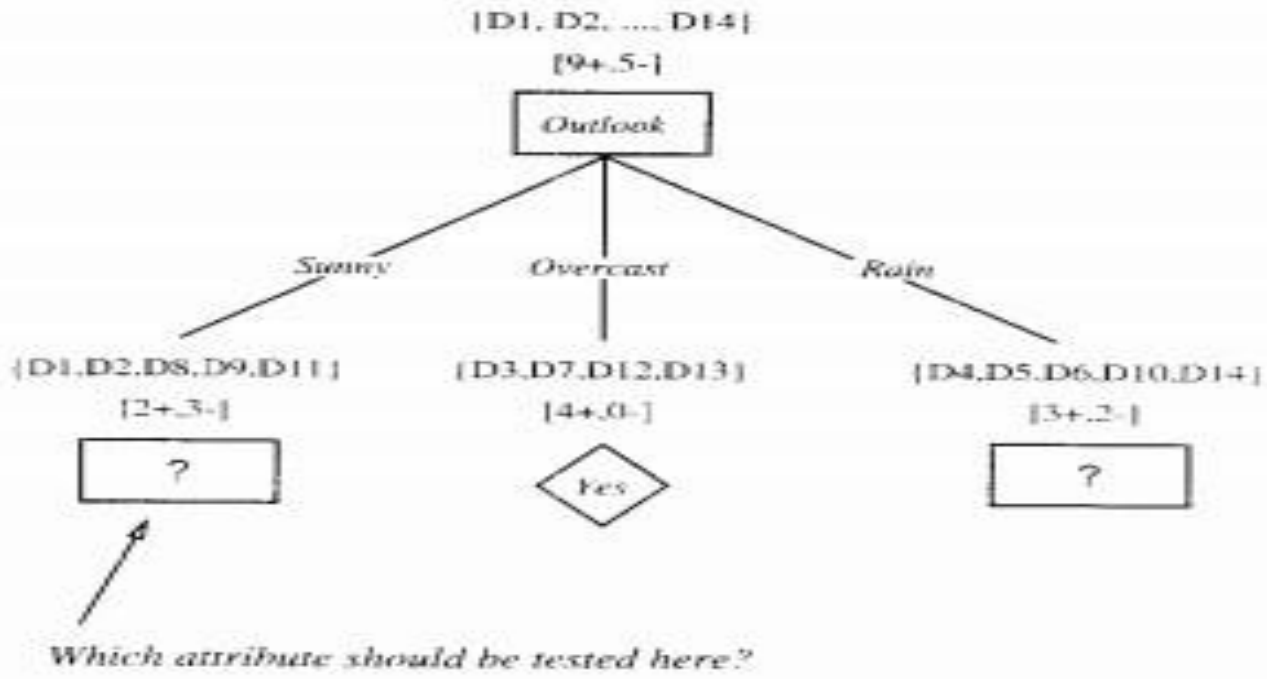
# Decision Tree (DT)

❖ The *partially learned decision* tree resulting from the first step of ID3




$S_{sunny}$ = {D1,D2,D8,D9,D11}

$Gain(S_{sunny}, Humidity) = .970 - (3/5)\,0.0 - (2/5)\,0.0 = .970$

$Gain(S_{sunny}, Temperature) = .970 - (2/5)\,0.0 - (2/5)\,1.0 - (1/5)\,0.0 = .570$

$Gain(S_{sunny}, Wind) = .970 - (2/5)\,1.0 - (3/5)\,.918 = .019$



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# Entropy of Humidity(partial tree)

- Values(Humidity)=High, Normal
- $S_{sunny}$=[2+,3-]  $S_{High}$□[3+,0-] $S_{Normal}$□[2+,0-]
- Entropy(High)= $3/3\log_2 3/3 - 0/3\log_2 0/3$

$= -(1)\log_2(1) - (0)\log_2(0)$

$= 0+0 = 0$

- Entropy(Strong)= $2/2\log_2 2/2 - 0/2\log_2 0/2$

$= (1)\log_2(1) - (0)\log_2(0)$

$= 0+0 = 0$

Gain(S, Humidity)=Entropy(S) – (3/5)Entropy(high) – (2/5)Entropy(normal)

$= 0.971 - (3/5)0 - (2/5)0$

$= 0.971$



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# Entropy of Temperature(partial tree)

- Values(Temperature)=Hot, Mild, Cool

$S_{sunny}$=[2+,3-] $S_{Hot}$□[2+,0-] $S_{Mild}$□[1+,1-] $S_{Cool}$□[1+,0-]

- Entropy(Hot)= $2/2\log_2 2/2 - 0/2\log_2 0/2$

$= -(1)\log_2(1) - (0)\log_2(0)$

$= 0+0 \quad = 0$

- Entropy(Mild)= $1/2\log_2 1/2 - 1/2\log_2 1/2$

$= (0.5)\log_2(0.5) - (0.5)\log_2(0.5)$

$= 0.5+0.5 \quad = 1$

- Entropy(Cool)= $1/1\log_2 1/1 - 0/1\log_2 0/1$

$= (1)\log_2(1) - (0)\log_2(0)$

$= 0+0 \quad = 0$

Gain($S_{sunny}$, Temp)=Entropy(S) – (2/5)Entropy(high)–(2/5)Entropy(normal) – (1/5)Entropy(normal)

$= 0.971 - (2/5)0 - (2/5)1 - (1/5)0$

$= 0.571$



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# Entropy of Wind(partial tree)

- Values(wind)=weak, strong
- $S_{sunny}$=[2+,3-]  $S_{weak}$□[1+,2-] $S_{strong}$□[1+,1-]
- Entropy(weak)= $1/3\log_2 1/3 - 2/3\log_2 2/3$

$= (0.333)\log_2(0.333) - (0.667)\log_2(0.667)$

$= 0.528+0.389$  $= 0.917$

- Entropy(Strong)= $1/2\log_2 1/2 - 1/2\log_2 1/2$

$= (0.5)\log_2(0.5) - (0.5)\log_2(0.5)$

$= 0.5+0.5$  $= 1$

Gain($S_{sunny}$, Wind)=Entropy(S) – (3/5)Entropy(weak) – (2/5)Entropy(strong)

$= 0.940 - (3/5)0.917 - (2/5)1$

$= 0.01$



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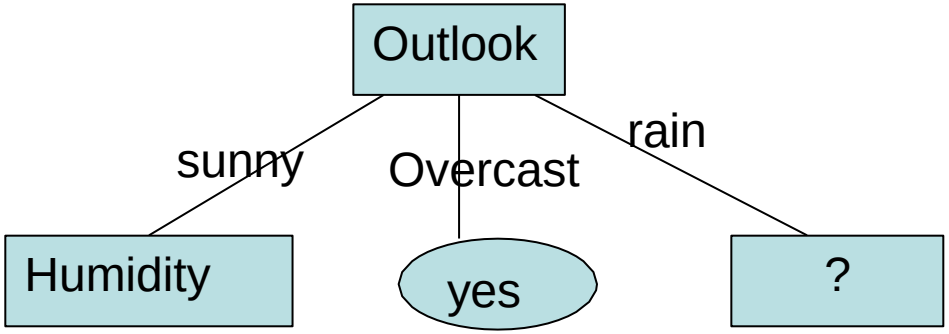
- Gain($S_{sunny}$, *Humidity*) = 0.971
- Gain($S_{sunny}$, *Wind*) = 0.01
- Gain($S_{sunny}$, *Temperature*) = 0.571



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# Decision Tree (DT)

- The *overcast* descendant has only positive examples and therefore becomes a leaf node with *classification Yes*:

- The other two nodes will be expand by selecting the attribute with the *highest information gain* relative to the new *subsets*.



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# Decision Tree (DT)

- Decision Tree learning can be:
  - *Classification* tree: finite set values target variables
  - *Regression* tree: continuous values target variable

- There are many specific DT algorithms:
  - ID3 (Iterative of ID3)
  - C4.5 (Successor of ID3)
  - CART(classification and Regression Tree)
  - CHAID (Chi – squared Automatic Interaction Detector)
  - MARS: extends DT to handle numerical data better



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# Decision Tree (DT)

- Different DT algorithms use different metrics for measuring the "*best attribute*" :
  - *Information gain*: used by ID3, C4.5 and C5.0
  - *Gini impurity*: used by CART



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# Decision Tree (DT)

- ID3 in terms of its search *space* and search *strategy*:
  - ID3's hypothesis space of all decision tree is a *complete* space of finite discrete-valued functions.
  - ID3's maintains *only a single current hypothesis* as it searches through the space of decision trees.

  - ID3 in its pure form perform *no backtracking* in its search. (post-pruning the decision tree)
  - ID3 uses all *training examples* at each step in the search to make statistically based decisions regarding how to refine its current hypothesis. (*much less* sensitive to *error*)



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# Decision Tree (DT)

- Inductive bias in DT learning (ID3):
  - Inductive bias are the set of *assumption*.
  - ID3 selects in favor of *shorter trees* over *longer* ones. (breadth first approach)
  - Selects trees that place the attributes with the *highest information* gain *closest to the root*.



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# Decision Tree (DT)

- *Issues* in DT learning:
  - How *deeply* to grow the decision tree
  - Handling *continuous attributes*
  - Choosing an *appropriate attributes* selection measure
  - Handling training data with *missing attribute values*
  - Handling attributes with *differing costs* and
  - Improve *computational* efficiency

- ID3 *extended* to address most of these issues to C4.5.



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# Decision Tree (DT)

- Avoiding over fitting the Data:
  - *Noisy* data and *too small* training examples are problems.
  - *Over fitting* is a practical problem for DT and many of the *learning algorithms*.
  - Over fitting was found to *decrease* the accuracy of the learned tree by *10-25%*.
  - Approach to avoid over fitting:
    - *Stop* growing the tree, before it over fitting. (direct but less practical)
    - Allow the tree to *overfitting*, and then *post-prune* (the most successful in practice)



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# Decision Tree (DT)

- Incorporating continuous-value attributes:
  - Initial definition of ID3, attributes and target value must be *discrete set* of values.
  - The *attributes tested* in the decision nodes of the tree must be *discrete value* :
    - Create a new *Boolean attribute* for the *continuous value* or
    - *Multiple interval* rather than just two interval.



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# Decision Tree (DT)

- Handling training example with missing attribute value:
  - Assign it with the *most common value* among training examples at node n.
  - Assign the *probability* to each of the possible values of the attribute .
  - The 2nd approach, is used in C4.5.



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# Question & Answer

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# Thank You !!!



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# Assignment I

❖ Answer the given questions by considering the following set of *training examples*.

| Instance | Classification | $a_1$ | $a_2$ |
|---|---|---|---|
| 1 | + | T | T |
| 2 | + | T | T |
| 3 | − | T | F |
| 4 | + | F | F |
| 5 | − | F | T |
| 6 | − | F | T |



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# Assignment I

- (a) What is the *entropy* of this collection of training examples with respect to the *target function classification*?
- (b) What is the information gain of $a_2$ relative to these training examples?



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# Decision Tree (DT)

- Select one *DT algorithms from the list below* and write an essay
  - ID3 (Iterative of ID3)
  - C4.5 (Successor of ID3)
  - CART(classification and Regression Tree)
  - CHAID (Chi – squared Automatic Interaction Detector)
  - MARS: extends DT to handle numerical data better



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